OPERATIONS RESEARCH 76000093 · FINAL PROJECT

Right Place, Right Time

Capacity, guidance and pricing for public EV charging in Shenzhen: five OR methods on 17,532 real charging piles

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Data: UrbanEV (Li et al., Scientific Data 2025), 275 traffic zones, hourly, Sep 2022 – Feb 2023

THE PROBLEM

Enough chargers, but in the wrong places and at the wrong times

28%

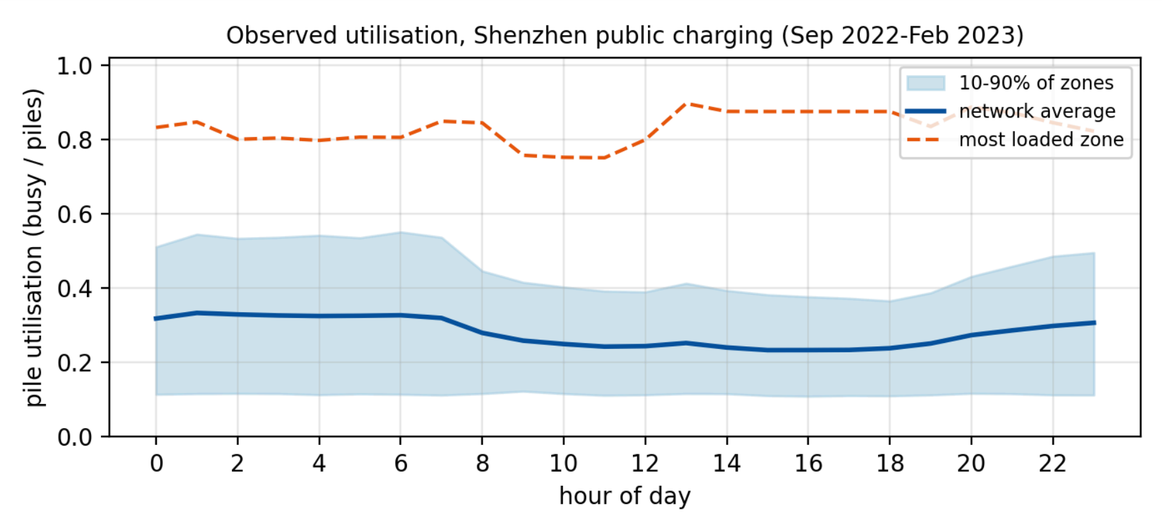
average occupancy of 17,532 public piles

10 zones

100% full on busy days (night peak, cheap tariff)

48 zones

unstable queues if demand doubles, while 44% of piles still idle



Night peak (0–6 h). Shenzhen had 1.08 M NEVs by Jun 2024; China's NEV stock grew 49% in the following year.

Sources: UrbanEV dataset; CnEVPost (Jul 2024, Jul 2025) · presented by Nguyen Van Thanh

OUR APPROACH

Five levers, five OR methods, one yardstick

M1

Queueing

Who waits, where and when?

M2

Integer program

Build new or move idle piles?

M3

Network flow

Guide drivers to spare piles?

M4

Nonlinear program

Re-time demand with prices?

M5

Simulation + SA

Do the models hold in reality?

Yardstick minimum budget to keep the average wait under 15 seconds per driver, with demand ×1.5, ×2 and ×2.5

Each method was checked independently: textbook values, LP relaxation, network simplex, KKT residual, simulation vs Erlang-C

M1 · QUEUEING THEORY

Every zone is an M/M/c queue, every hour

Servers c = piles in the zone (4 to 373)

Load a = busy piles observed each hour, × growth g

Erlang-C P(wait) = C(c, a), Wq = C·τ / (c − a)

Daily cost W(c) = Σ over hours of λ·Wq

Erlang-B recursion keeps it stable up to 373 servers; checked against textbook values (C(2, 1.6) = 0.7111)

A USEFUL RESULT

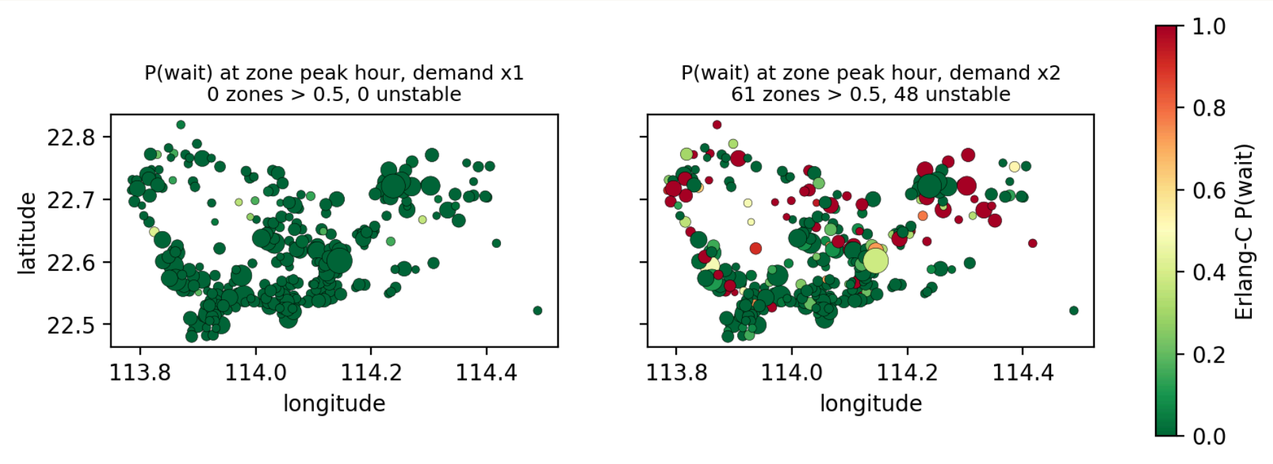
Total waiting does not depend on session length τ

$$\lambda \cdot Wq = (a/\tau) \cdot C \cdot \tau/(c - a) = a \cdot C/(c - a)$$

The data give busy piles, not sessions. This identity removes the unknown τ from every planning decision.

M1 · RESULTS

Doubling demand breaks 48 zones at 56% average use



demand	avg use	unstable
today (×1)	28%	0
×1.5	42%	13
×2 (base)	56%	48
×2.5	70%	91

A mismatch, not a shortage: at ×2, a waiting probability ≤ 20% in every zone needs 14,367 piles, fewer than the 17,532 that exist.

unstable = peak load reaches the number of piles

M2 · INTEGER PROGRAMMING

Build new or move idle piles? An exact MILP

min Σ Wz(c + add − remove)

s.t. new + 0.35 · relocated ≤ budget B

every zone stable: piles > peak load

donors give at most half of their piles

integer piles in every zone

Why it is exact

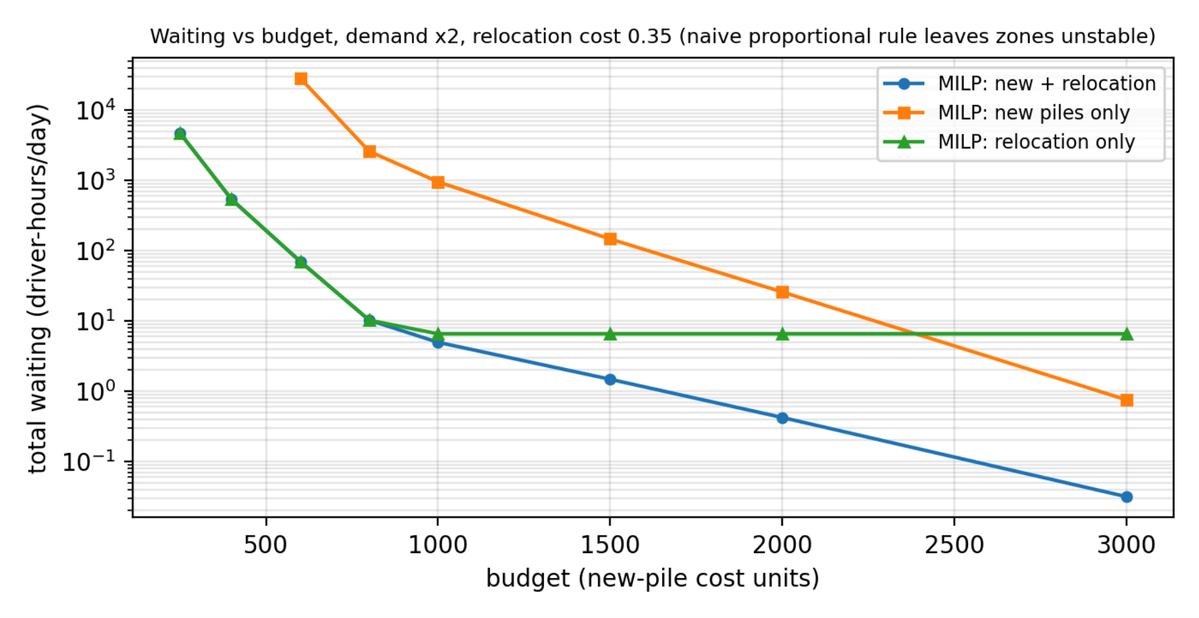
Waiting is convex in the number of piles (verified in all 275 zones), so each extra pile is a 0-1 increment with a smaller gain than the previous one.

About 8,400 binaries, solved by HiGHS in seconds; LP relaxation gap only 0.003 h/day.

relocation cost 0.35 of a new pile (sensitivity 0.1–1.0)

M2 · RESULTS (DEMAND ×2)

Moving idle piles beats building by 190×



5.0 vs 952

driver-h/day at budget 1,000: optimal mix vs new piles only

41 zones

still unstable with the naive proportional rule

105 → 0.01

shadow price (h/day per unit) from B = 250 to 1,000: steep diminishing returns

stability costs 209 units with relocation vs 598 with new piles only

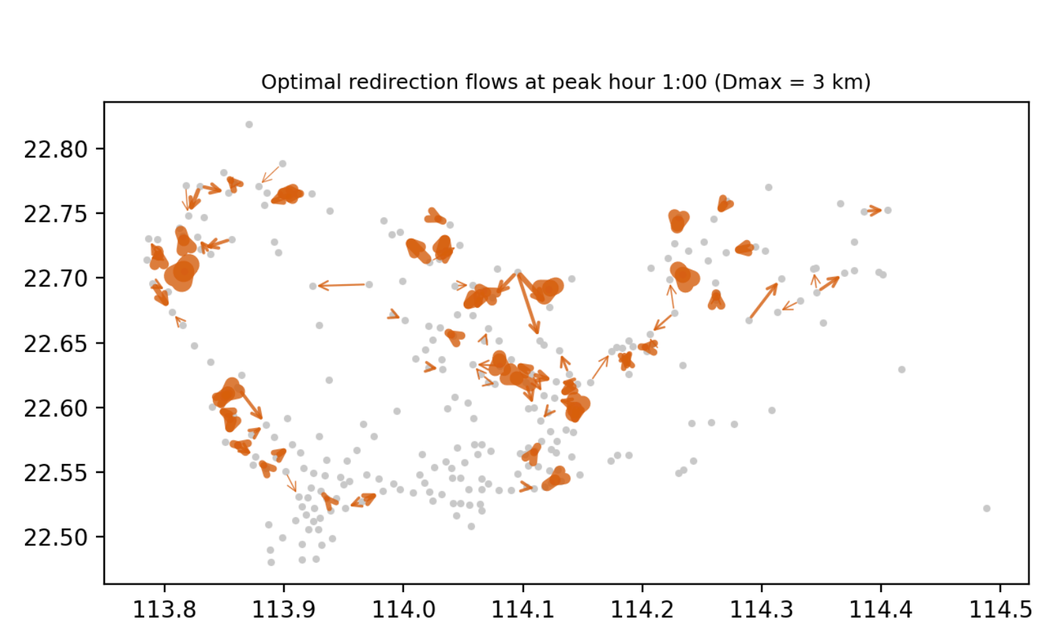
M3 · NETWORK OPTIMISATION

Guiding drivers 3 km removes 69% of the shortfall

Min-cost flow per hour: overloaded zones send excess load to zones with spare capacity within D km, minimising detour km

detour limit	redirected	piles needed
none	0%	598
2 km	50%	319
3 km	70%	185
5 km	95%	32

Max flow = min cut (748 at the peak hour) certifies the limit: 23 zones whose 3 km neighbourhood is saturated. That is where piles are needed.



target utilisation 0.85; mean detour 1.7 km at D = 3 km

M4 · NONLINEAR PROGRAMMING

A revenue-neutral time-of-use fee

1 · Behaviour

Drivers pick the charging hour by a logit on total price; daily demand is conserved

2 · Estimate β

118 TOU-priced vs 157 flat-priced zones, zone fixed effects: $\beta = 0.28$ ($|t| = 2.2$), elasticity ≈ -0.44

3 · Optimise

24 hourly fees minimise overload; same average fee revenue; KKT residual $< 10^{-7}$

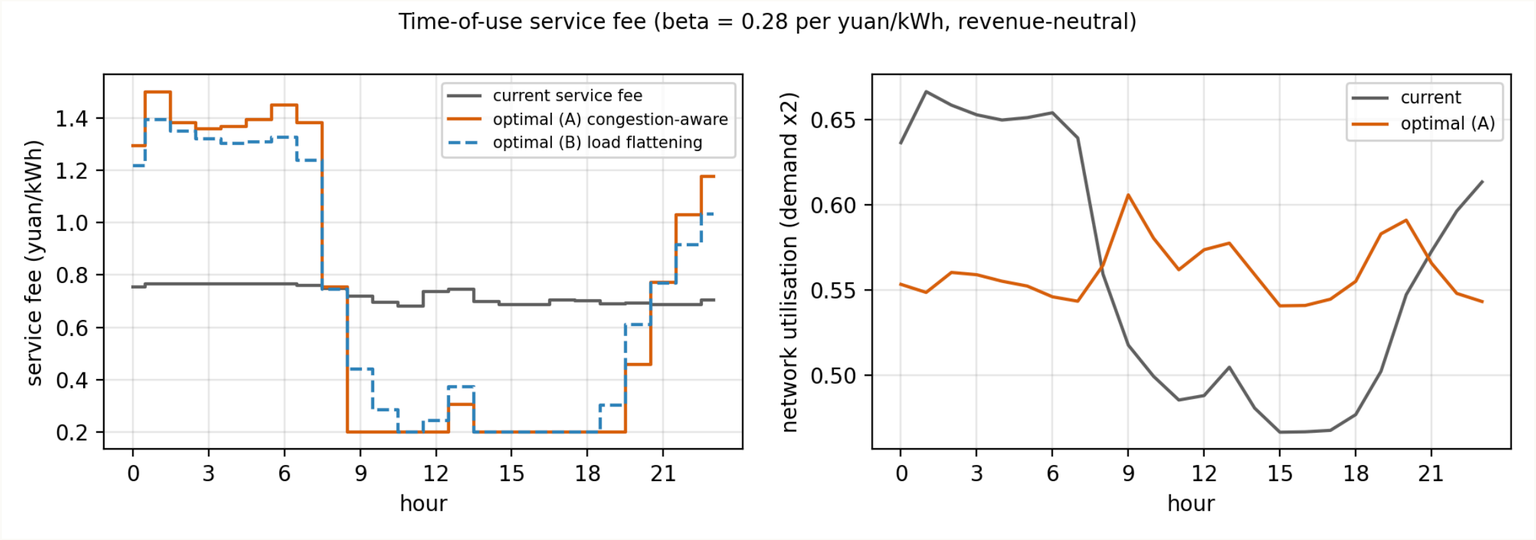
LESSON FROM A ZERO MULTIPLIER

With “revenue ≥ today” the constraint never bound ($\mu = 0$): since demand is conserved, fees could rise without limit. Revenue neutrality fixes the degenerate optimum; now $\mu = 0.19$ prices the revenue–congestion trade-off.

solved with trust-constr, cross-checked with SLSQP

M4 · RESULTS (DEMAND ×2)

Price fixes the “when”, not the “where”



−49%
congestion index

598 → 403
piles needed for stability

48 → 45
unstable zones: time lever, space problem

moderate fees 0.5–1.0 yuan/kWh: 598 → 467 piles · robust for β from 0.1 to 1.0

M5 · DISCRETE-EVENT SIMULATION

Do the analytical models hold in reality?

30-zone cluster around the worst zone, demand ×2, about 88,000 arrivals per run

Hourly-varying arrivals · log-normal sessions (CV 0.8)

Spill-over: 3 or more drivers queueing → divert to the nearest free zone within 3 km

Verified against Erlang-C

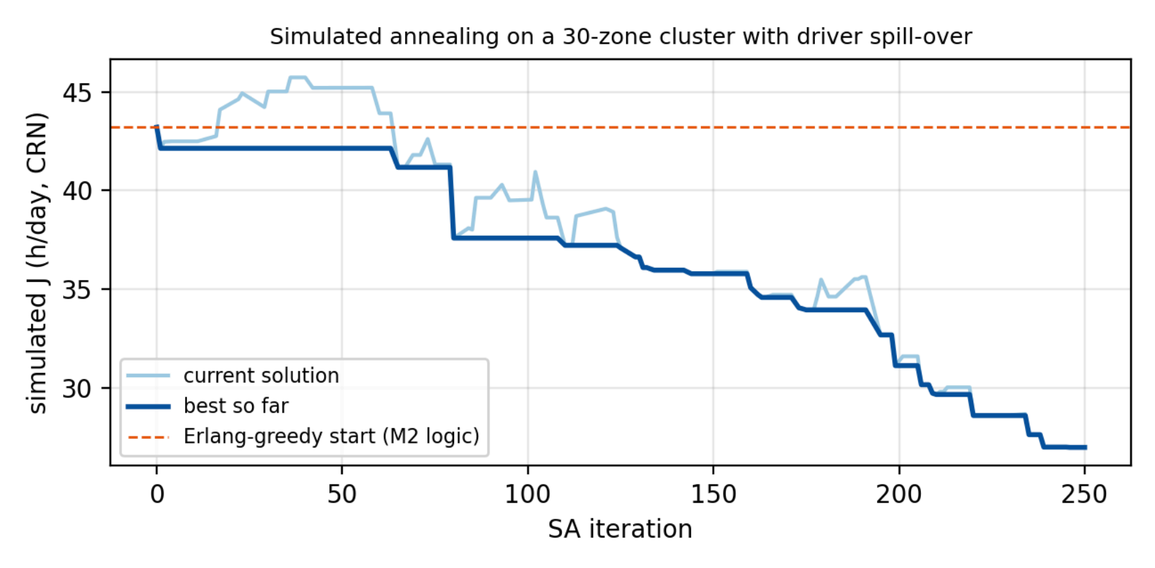
(c, a)	Erlang Wq	simulated ± s.e.
(5, 4)	33.2 min	34.3 ± 3.0
(20, 17)	7.7 min	7.8 ± 0.5
(50, 45)	4.4 min	4.4 ± 0.2

Erlang (SIPP) predicts 203 waiting h/day for the M2-type plan; simulation gives 95. Conservative, because peaks are too short for queues to reach steady state.

60 days × 8 replications for verification

M5 · SIMULATED ANNEALING

Spill-over-aware search: 34% better than the Erlang plan



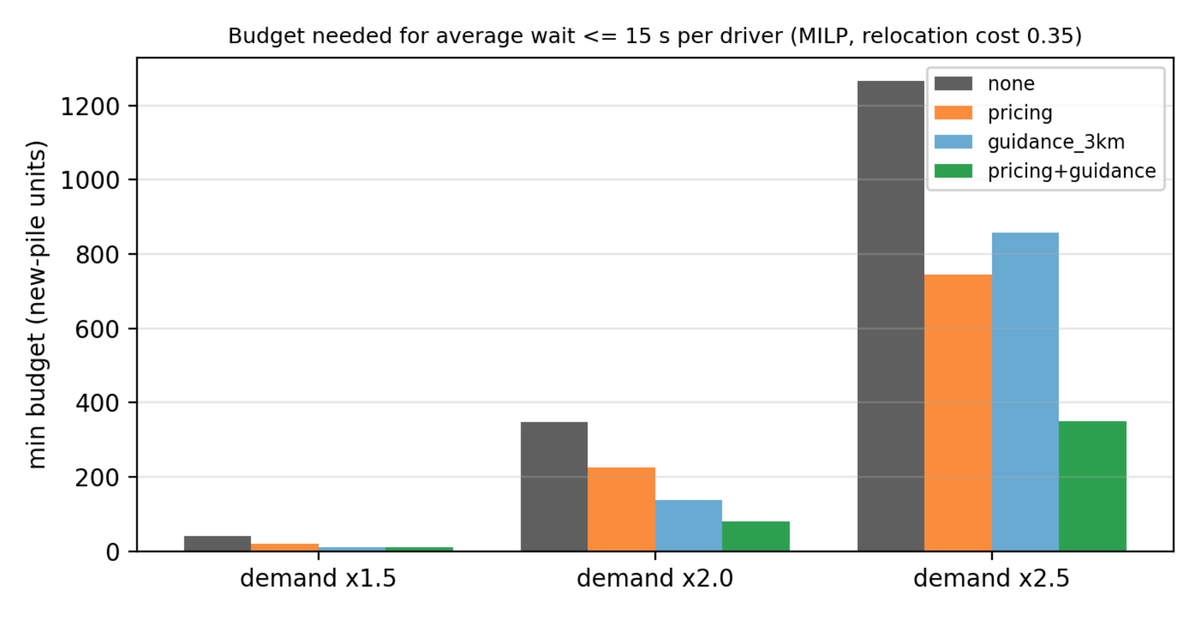
out of sample	Erlang plan	SA plan
cost J (h/day)	46.8	31.0
waiting (h/day)	41.5	23.6
detour (km/day)	107	148

−15.8 ± 3.6 h/day (95% CI) by moving 21 of 174 piles; still 32% with stability enforced. Better under every diversion setting tested (10–43%), but worse if drivers never divert: deploy with guidance.

common random numbers during search; J = waiting + 0.05 h per detour km

PUTTING IT TOGETHER

Soft levers cut the budget needed by 70–77%



demand ×2	min budget
no soft lever	348
pricing	226
guidance 3 km	138
pricing + guidance	81

At demand ×2.5: 1,266 → 350. Time and space levers are complements: together they beat either alone.

Budget in new-pile units (relocation 0.35) for an average wait ≤ 15 s per driver; MILP with bisection on B

RECOMMENDATIONS

Software first, hardware last

1. **Guide** drivers to free piles within 3 km in navigation apps: two thirds of the shortfall, almost no hardware
2. **Relocate** idle central piles to the growing north-west and north-east, following the MILP plan
3. **Price** with a moderate, revenue-neutral night/day fee to flatten the night peak
4. **Build** only where duals are binding; stop when the budget's shadow price falls below the value of drivers' time
5. **Re-plan yearly**: required piles grow far faster than demand (61 → 598 → 1,931 at ×1.5 → ×2 → ×2.5)

All five members

LIMITATIONS AND NEXT STEPS

- Occupancy is carried load: unserved demand at full zones is invisible
- Uniform growth: zone-level forecasts would sharpen the plan
- β weakly identified: a tariff experiment would help
- Next: grid limits, stochastic two-stage planning, bilevel pricing

Thank you. Questions?

Right place, right time: guide, relocate, price, then build.

Code, data and results in the project ZIP · UrbanEV data (CC0)